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Module 2 Introduction

Lesson 1: Linear Regression and Least Squares

Lesson 2: Modification of Linear Regression: Subset Selection

Lesson 3: Coefficient Shrinkage for Linear Regression: Ridge Regression and LASSO

Reading: Coefficient Shrinkage for Linear Regression: Ridge Regression and LASSO Readings 20 min

Video: Ridge Regression 10 min

Video: LASSO 10 min

Practice Assignment: Coefficient Shrinkage for Linear Regression: Ridge Regression and LASSO Quiz Submitted

Lab: Coding Exercise 10 min

Lesson 4: Data Transformations and Linear Regression

Module 2 Summative Assessment

Module 2 Summary

Your grade: 100%

Your best: 100% • Your highest: 100%

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Coefficient Shrinkage for Linear Regression: Ridge Regression and LASSO Quiz

Instructions

1. What is the primary difference between Ridge Regression and LASSO regarding the regularization term they add to the loss function? Review Learning Objectives 1 / 1 point

☐

Ridge Regression adds the absolute magnitude of coefficients as the penalty term, while LASSO adds a squared magnitude.

☐

Ridge Regression adds the squared magnitude of coefficients as the penalty term, while LASSO adds a squared magnitude.

☒

Ridge Regression adds the squared magnitude of coefficients as the penalty term, while LASSO adds the absolute magnitude.

☐

Both add the same type of penalty term to the loss function.

☐

Neither adds any penalty term to the loss function.

Correct

Assignment details

Submitted Nov 12, 2024 10:57

Attempts Unlimited

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2. Which of the following is an advantage of LASSO over Ridge Regression when it comes to feature selection? 1 / 1 point

☐

LASSO does not allow for the selection of features.

☐

LASSO is computationally more efficient than Ridge Regression.

☐

LASSO can only shrink coefficients of the model.

☒

LASSO can result in sparse models where some coefficients are exactly zero.

Correct

Assignment details

Submitted Nov 12, 2024 11:02

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3. In the context of Ridge Regression, what is the effect of a very large value of the regularization parameter, lambda? 1 / 1 point

☐

The model will become equivalent to a simple linear regression.

☐

The coefficients will be very large.

☐

The coefficients will become exactly zero.

☒

The coefficients will shrink towards zero, but not become exactly zero.

Correct

Assignment details

Submitted Nov 12, 2024 11:05

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